

Reference sample — written for the BlueWave Telecom scenario, not your Sankofa Digital case. The structure, length, and writing style are what to copy. The findings are not.

Stage 0 Capstone · Deliverable 1

Suspicious-login evidence table

Author: Janet Mukami Kariuki **Intern code:** UBI-2026-DEMO **Date completed:** 02 June 2026 **Programme:** Ubuntu Bridge Initiative · Cybersecurity Internship · Cohort 1 **Stage:** 0 · Induction at the Gate

Methodology

I read `partner-portal.log` first to build a chronological view of authentication events between 12 November and 4 December 2023, then cross-referenced every suspicious line against `change-tickets-q4.csv` to determine which had been raised by the SIEM and how Daniel Otieno (the responding Tier-1 analyst at the time) had dispositioned each. I then loaded `vendor-roster.csv` to confirm which vendor accounts were active versus terminated during Q4. The four encoded payloads in `encoded-blob.txt` were decoded last; payload 3 resolved to a threat-actor name that I then back-searched in `partner-portal.log`.

A login was treated as suspicious if at least one of the following held: the source IP did not match the vendor's documented network range (per `vendor-roster.csv`), the account belonged to a roster entry marked terminated, the access fell outside the contracted change window, or the SIEM rule that fired correlated to two or more prior alerts in the same week. Where the log and the ticket history disagreed, I trusted the log timestamp and treated the ticket disposition as the analyst's interpretation, not as ground truth. Confidence ratings were assigned per the definitions below.

Confidence rating definitions

H — High confidence. Two or more independent files corroborate the event; no plausible benign explanation exists.

Example: row 1 — the BWT-29104 SSO acceptance is corroborated by `partner-portal.log:142`` and `vendor-roster.csv:18`` (`vendor-ace.ke`` terminated 22 October 2023).

M — Medium confidence. One file shows the event directly; corroboration is plausible but not confirmed within this pack.

Example: row 5 — the second IP attempt at `169.150.224.12`` against the same terminated account is confirmed in `partner-portal.log:412`` but the SIEM dispatch in `change-tickets-q4.csv`` is missing the cross-reference.

L — Low confidence. Suspicious by pattern only; no single line of evidence directly confirms it.

Example: row 8 — the off-hours OAuth token refresh on 02 December 23:47 UTC matches the broader pattern but appears in only one file.

Suspicious-login events, Q4 2023 (BlueWave partner portal)

#	Timestamp UTC	Source IP	Account	Why suspicious	Corroborating evidence ("e 2 files)	Confidence
1	2023-11-14 23:41	102.215.13.4	vendor-ace.ke	SSO acceptance on terminated account from a non-vendor IP	partner-portal.log:142; vendor-roster.csv:18; BWT-29104	H

2	2023-11-19 22:08	102.215.13.4	vendor-ace.ke	Repeat from same IP, same account, five days later	partner-portal.log:218; change-tickets-q4.csv:24 (BWT-29178)	H
3	2023-11-21 03:14	102.215.13.4	vendor-ace.ke	OAuth refresh-token acceptance immediately followed by a billing-CSV export request	partner-portal.log:251–263; change-tickets-q4.csv:29 (BWT-29222)	H
4	2023-11-22 00:02	102.215.13.4	hyena-collective	Threat-actor name appears in payload-3 of `encoded-blob.txt`, then in the same session as row 3	encoded-blob.txt payload 3; partner-portal.log:271	H
5	2023-12-03 23:58	169.150.224.12	vendor-ace.ke	Second source IP against the same terminated account; SIEM hit but no ticket cross-reference	partner-portal.log:412; vendor-roster.csv:18	M
6	2023-11-26 14:42	102.215.13.4	vendor-ace.ke	Daytime access during Ace's documented change window; checked against contracted calendar — Ace's contract was already terminated 22 Oct	partner-portal.log:339; vendor-roster.csv:18; vendor-contract-2023.json	M
7	2023-11-29 11:21	102.215.13.4	vendor-noor.bw	Same source IP rotating to a different vendor account — lateral pivot attempt	partner-portal.log:381; vendor-roster.csv:34	M

8	2023-12-02 23:47	102.215.13.4	vendor-ace.ke	OAuth refresh-token issuance outside any logged interactive session — automated, not human	partner- portal.log:401	L
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Per-row commentary

Row 1. This is the breach itself — the moment the attacker first had legitimate-looking access. The IP 102.215.13.4 resolves to a Nairobi residential ISP that is not in Ace's documented network range, and `vendor-ace.ke` was offboarded 23 days earlier. Daniel closed BWT-29104 as "vendor activity — expected", which becomes the dismissal pattern D2 unpacks. This is the row I would reread first.

Row 2. A repeat from the same IP and account five days later proves the attacker bookmarked the access. The interval suggests planned return, not opportunistic re-entry. BWT-29178 inherited BWT-29104's disposition ("Same as BWT-29104. No new info."), validating the pattern and hardening it for subsequent alerts.

Row 3. This is the row where the attacker's behaviour shifted from lurking to action. Within nine minutes of acceptance, an OAuth refresh-token was minted and a billing-CSV export was requested. Customer data left the building between 03:14 and 03:23 UTC. Everything before this row is reversible; everything after it is breach-notification territory.

Row 4. The decoded payload-3 of `encoded-blob.txt` resolves to "Hyena Collective". Payloads 1, 2, and 4 decode to operational instructions (target list, schedule, exfil destination); only payload 3 is the actor self-identifier. The same string appears in `partner-portal.log:271` as a user-agent fragment in the row-3 session.

Row 5. A second source IP (169.150.224.12) against the same terminated account three weeks later suggests the attacker uses a small pool of residential ISPs and rotates between them. The SIEM raised this but no ticket cross-referenced it to the earlier four events.

Row 6. Daytime access on 26 November during Ace's normal change window looks routine until you note Ace's contract was terminated on 22 October — they have no change window. The vendor calendar would have flagged this if cross-referenced.

Row 7. On 29 November the same source IP rotated to a different vendor account (`vendor-noor.bw`). This is the moment we know the attacker had more than one credential — and that they are systematically working through BlueWave's vendor list.

Row 8. An OAuth refresh-token issuance at 23:47 with no logged interactive session above it suggests automated activity — a script holding the credentials, not a human typing them. This is L confidence because the inference depends on the absence of a session, not on the presence of evidence.

Mandatory rows that prove the case

- **Row 1** is the breach: stolen vendor SSO on a terminated account, BWT-29104, 14 November.
- **Row 2** is the planned repeat: BWT-29178, 19 November.
- **Row 4** names the threat actor as "Hyena Collective" via `encoded-blob.txt` payload 3.
- **Row 1** also satisfies the post-offboarding-access requirement: `vendor-ace.ke` was terminated 22 October per `vendor-roster.csv:18`.

Pattern summary

The eight rows above describe a single attacker (one primary IP, two related accounts, one decoded threat-actor name) returning at least seven times across twenty days to the BlueWave partner portal, escalating from authentication (rows 1–2) to OAuth-token theft and data exfiltration (row 3) to lateral attempts against other vendor accounts (row 7) to automated token refresh (row 8). Three of the seven events were raised by the SIEM and three of the seven were dismissed by the same Tier-1 analyst as "vendor activity — expected". The dismissal pattern across rows 1, 2, and 6 is the throughline that D2 unpacks.

The row a grader should reread first

Row 3. It is the only row where the attacker's behaviour transitioned from "lurking" (authentication) to "doing

damage" (OAuth-token theft followed by a customer-data export within nine minutes). Everything before row 3 is reversible; everything after it is breach-notification territory and a 72-hour regulator clock. If only one piece of evidence makes it into the Incident Committee report, it is this row.

My next investigative step

Pull the full OAuth token-issuance log for `vendor-ace.ke` between 12 November and 4 December and correlate refresh-token timestamps against the four partner-portal logins above. If the refresh tokens were minted on dates that do not align with vendor-portal logins, that confirms a secondary compromise vector outside what this pack shows — most likely the IdP itself. The query is one Splunk search against `idp.oauth_tokens` keyed by `subject=vendor-ace.ke`.

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